One thing I have learned throughout my master's is that the most interesting machine learning problems often emerge when the data itself does not fit neatly into existing modelling assumptions. My current thesis began as a content moderation problem but quickly became a multimodal learning problem involving text, images, GIFs, and contextual information. Understanding how to represent these different sources of information in a useful way became more important than the choice of model itself. That experience is one of the reasons this PhD project appeals to me. Microscopy images are a very different domain, but the challenge is learning meaningful representations from complex, high-dimensional data. The prospect of using those representations to generate biological insight makes the problem even more compelling.

I recently completed my Master's degree in Data Science at Uppsala University after previously studying Software Engineering. Throughout my master's, I found myself increasingly drawn toward research problems that sit between machine learning and healthcare. Through coursework and projects, I explored healthcare innovation, data interoperability, and the broader challenges involved in turning complex biomedical data into useful knowledge. These experiences reinforced my interest in working on research that combines methodological development with practical scientific questions.

My strongest technical experience comes from machine learning, software development, and working with complex datasets. Through my thesis, I have gained experience designing and evaluating custom deep learning architectures, handling multimodal data, and conducting independent research under considerable uncertainty. Beyond the technical aspects, I enjoy the process of understanding a problem deeply enough to decide what information is actually useful before attempting to model it.

I do not have a formal background in computational biology or microscopy image analysis. However, that is precisely part of what attracts me to the position. The opportunity to work at the intersection of computer vision, representation learning, and biology would allow me to develop new domain expertise while contributing my existing strengths in machine learning and software engineering. I am particularly interested in how the AMBIOM group connects methodological advances to biological insight, and how computational models can help reveal structures and patterns that would otherwise remain hidden in large-scale imaging data.